Remarks on Rix's Law in Greek

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The phonological development known as "Rix's Law" (Rix 1970) describes the treatment in Greek of initial laryngeals before pre-consonantal syllabic resonants. The status of this process remains somewhat uncertain: Rix's proposal has met with both a large measure of acceptance as well as strenuous objection, the latter most notably (and repeatedly) from F. O. Lindeman, among other scholars (e.g. Meißner 1998a). There may well be reason to reject the subpart of Rix's Law that treats the environment before syllabic nasals (Nikolaev 2004). But on the basis of new arguments and hitherto unconsidered evidence (mainly from the Greek dialects and Mycenaean), it is shown that for the context before syllabic liquids, the counterarguments of Lindeman and others are without value, and that therefore Rix's Law — one of the cornerstones of the "triple representation" theory of laryngeal development in Greek — must be maintained for that environment

1. In his seminal paper on "Anlautender Laryngal vor Liquida oder Nasalis sonans im Griechischen", published in 1970, H. Rix sought to demonstrate for Greek a development whereby initial laryngeals before pre-consonantal syllabic resonants were replaced by their "vocalized" counterparts (exactly as with pre-consonantal laryngeals that have vocalized as so-called "prothetic vowels"), while the resonants themselves ended up as non-syllabic. Thus, in Rix's words (1970:80): "Man kann als Lautgesetz etwa formulieren: Für idg. h_1- h_2-h_3- im Anlaut vor r̥ l̥ m̥ n̥ erscheint im Griechischen ἐ- ἀ- ὀ- vor r l m n (*/e a o/ + /r l m n/*)".[1] This proposed sound law, now known as "Rix's Law" (or "Lex Rix"),[2] has been widely accepted, at least in its original form,[3] to judge from various

[1]For the laryngeals, I have substituted H-notations for Rix's original notations with <ə>.

[2]For the "canonization", see e.g. Collinge 1985:236f.

[3]I leave aside the problem of possible applications to Latin (see already Rix 1970:96), which have been more controversial (references at Meier-Brügger 2003:123, see also Schrijver 1991:56ff., Meiser 1998:106, Lindeman

influential handbook or handbook-style treatments,[4] from other theoretical discussions,[5] and from subsequent applications to other Greek material adduced in support of the process.[6]

Nevertheless, strenuous objections to Rix's Law have been repeatedly voiced in the literature, most notably by F. O. Lindeman,[7] an eminent specialist in laryngeal theory.[8] Nor has Lindeman stood alone in his objections: these have been echoed forcefully by W. Meid (1988:344f. with n. 14), and more recently by T. Meißner, in an extended treatment (1998a).[9] There is thus some interest in reassessing the problem, especially since — as is my contention here, in taking up Lindeman's challenge — it may be possible to adduce some new arguments and evidence bearing on Rix's Law that have never been considered, either by its supporters or critics.

2.0. The counterarguments to Rix's Law raised by Lindeman and others — most of them astutely anticipated by Rix himself — are essentially of four types (with some inevitable overlap), as follows.

2.1. *Disputed etymologies.* In his original presentation, Rix argued in favor of aligning Gk. ἔρχομαι 'go' with Ved. *ṛcháti* 'goes, moves' (and associated material in Old Iranian), under a preform **h₁ŕ-ské-* (1970:98ff.). Such a reconstruction is impeccable for the Indo-Iranian forms, both phonologically and morphologically; on the Greek side, the reconstruction

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2003a:96n2), as well as similar phenomena elsewhere (e.g. Peters 1993:398n105 on Macedonian, Hackstein 1998:217ff. on Tocharian).

[4]E.g. Mayrhofer 1986:129f., Beekes 1988:71, Meier-Brügger 2003:122f.

[5]E.g. Rasmussen 1983:72, 1990:449, Ringe 1988:429ff.

[6]The most elaborate of these is Peters 1993 on non-Att. ἐρσήν (discussed below, **3.5.**).

[7]See especially 1982:60ff., 1987:43, 1989:271f., 1990:19, 1994, 1997:54ff., 2003b:304n12, including a challenge: "Since I see no reason to withdraw my earlier contention, I challenge all those who endorse 'Lex Rix' to enlighten me on this subject by stating why they regard my criticism of 'Lex Rix' as unjustified" (1989:272).

[8]Note, however, that Lindeman's broader opposition to the "triple representation" of the laryngeals in Greek (see especially Lindeman 1982, and most recently 2004) renders Rix's Law antithetical *ipso facto* to this fundamental tenet of Lindeman's version of laryngeal theory.

[9]I am grateful to Joshua Katz for bringing Meißner's work to my attention.

depends, in addition to Rix's Law, on the relatively uncontroversial development of pre-Greek **-rsk-* to *-ρχ-*.[10] According to Lindeman, however, "Gk. ἔρχεται, which is frequently connected with an IE. root **ser-*, has no established etymology" (1982:61).[11] Sub-types of this form of criticism include cases where a word with an accepted etymology apparently fails to undergo Rix's Law, counter to expectations,[12] and cases in which doubts are raised as to whether a given root whose Greek avatar is adduced in support of Rix's Law actually contained an initial laryngeal at the PIE stage.[13]

In the face of such objections, one can of course attempt to refute them, as has already been done to some extent.[14] In the end, however, Rix's Law must stand or fall on the basis of a residue of cases that lie beyond dispute, with respect to the features just described. Thus, even the critics grant that the first compound member ἀργι- 'bright-' has an unassailable etymological background, including an initial laryngeal that is

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[10]On this development, see Rix's discussion (1970:99), as well as that of Klingenschmitt (1974:274f.).

[11]Similarly and in more detail Meid 1988:345n14 ("... in der Mehrzahl die Beispiele entweder inkonklusiv sind oder sich nicht auf sichere Etymologie stützen können", and further "Die Etymologien sind nicht vom besten"), noting specifically that ἔρχομαι could be interpreted equally well as a full-grade **(h$_1$)érĝ^h-e-*, with a stem directly comparable to that seen in forms like OIr. (imper.) *eirgg, erggid* (suppletive to *tíagu* 'I go'). The treatment in LIV (238, s.v. **h$_1$er-*) favors Rix's analysis, but acknowledges the same ambiguity ("kann aber auch zu **h$_1$erĝ^h-* gestellt werden; in ἐρχε- könnten **h$_{1ĝ}$r-ské-* und thematisiertes **h$_{1ĝ}$rĝ^h-e-* zusammengefallen sein", 238n8).

[12]The clearest case of this sort is 'duck', discussed in these terms by Lindeman at 1990:19 (zero grade **h$_2$u̯h$_2$-tíh$_2$* [cf. Ved. *ātí-* 'type of water fowl'] > Att.-Ion. νῆττα/νῆσσα, (pseudo-)Boeot. νᾶσσα, and not a form of the type †/anassa/, as would be predicted by Rix's Law; similarly Meißner 1998a:39f.).

[13]Thus Meißner: "the postulation of a laryngeal for the root found e.g. in dat. sg. ἀλκί 'strength' < **h$_{2ĝ}$lk-i* is gratuitous and even questionable since there is no trace of it in Indo-Iranian, cf. Skt. ipf. *árakṣa-*" (1998:38).

[14]The problem with 'duck', for example, was subsequently addressed by Rix 1991 (whose arguments are nevertheless rejected by Lindeman [1994:43ff., 1997:159n187] and by Meißner [1998a:40n17]) and (with an important modification of Rix's general approach) by Katz 2004:197ff. (cf. also Rasmussen 1978:35, 1983:67n2, operating with initial **a-* instead of **h$_2$-*); on the laryngeal in ἀλκί, see further below (**3.2.**).

guaranteed by the Hittite cognate *ḫarki-* 'white' (most explicitly Meißner 1998a:38).[15]

2.2. *Disputed morphology.* Since Rix's Law applies to zero-grade forms, this objection claims that at least some alleged Rix's Law forms are in fact full grades, or have been secondarily influenced by full grades. Such a claim, moreover, is abetted by the fact that full-grade sequences of the type **HeRC-* would produce the same phonological result predicted by Rix's Law for zero-grade sequences, e.g. **h₂eRC-* and **h₂R̥C-* both give /aRC-/.[16] The problem here is real enough: especially in view of the potential for phonological ambiguity just noted, it is often not easy to guarantee that a given form is actually best interpreted as continuing a zero grade, as opposed to a full grade. Nevertheless, it is difficult to follow Meißner in his extreme assertion (1998a:38) that "every single case is explicable in this way" (scil. in theory as a full grade). Thus, there seems in fact to be no disagreement that ἀργι- (**2.1.**) is most likely to continue a zero grade **h₂r̥ĝ-i-* (cf. Ved. *r̥jí°*, Av. *ərəzi°*), however one wishes to assess the ultimate morphological quiddity of such "Caland" forms (cf. recently Meißner 1998b, Rasmussen 2002:335ff., 2004:275).[17] In a morphologically transparent case of this sort, naysayers must resort to claiming that the original zero grade has been influenced secondarily by one or more related full grades, as attested, for this root, in Greek forms like ἄργυρος 'silver', ἐναργής 'distinct, visible'. But this is not of itself an attractive hypothesis for ἀργι- (or for ἀργός, in its traditional interpretation), as Rix had already noted: "eine Übernahme der Vollstufe von diesen Wörtern [scil. ἄργυρος, ἐναργής] ist aber wegen der unterschiedlichen Bedeutung wenig wahrscheinlich" (1970:85).

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[15]There is also general agreement on the pendant adjective ἀργός 'bright, quick', traditionally considered to reflect a **-ro-* adjective **ἀργρός (cf. Ved. *r̥jrá-*, Av. *ərəzra-* 'brilliant, quick'), with dissimilatory loss of the second liquid; but this case is actually ambiguous, more on which below (**2.2.** with n. 17).

[16]For discussion of various Rix's Law instances in these terms, see especially Lindeman 1982:61, 1987:43, 1997:55f., Meid 1988:345n14, Meißner 1998:38f.

[17]The traditional interpretation of ἀργός (n. 15 above) is in fact insecure, as the form may well reflect a full-grade **h₂erĝ-o-* (cf. **h₁reu̯dʰ-o-* 'red', another adjective with Caland system associations), on which see Rau 1998:159.

2.3. *Disputed phonology.* For the phonological interpretation of the process he was describing, Rix observed that at least for the position involved (i.e. #HR̥C-), the replacement of the laryngeals by their vocalized versions in Greek must have occurred before the development of the "Stützvokal" associated with the treatment of the syllabic resonants (> Gk. *a*R/R*a* or *o*R/R*o*, depending on dialect). To this, Lindeman countered that "Rix's assumptions ... are in conflict with what has hitherto been the communis opinio regarding the behaviour of the IE. laryngeals", which in prevocalic position (and thus also, presumably, in the position before syllabic resonants) should simply be lost outside of Anatolian (1982:60, similarly 1987:43, 1989:272, 1997:54f.). Lindeman goes on (most forcefully at 1989:272) to describe what seems to him a phonological conundrum posed by the laryngeal behavior that Rix claimed:

> if the starting point was *#[H_1r̥*ske-*], how, then, could the *consonantal* 'laryngeal' *H_1 be '*replaced by a prothetic vowel*' (i.e. *H_1- > *e-*) in the position before the vocalic resonant **r̥ in the same way* as an initial 'laryngeal' in the position before a following consonantal liquid or nasal (type *#[H_2*ner-*] > Gk. *aner-*)? As the phonetic environment of *#[*HR̥C-*] differs from that of *#[*HRV-*], there is no reason whatever to expect a priori the same development of the inital 'laryngeal' in these cases.[18]

These observations deserve to be considered seriously, but they do not of themselves actually refute Rix's Law — they merely highlight certain limitations in our understanding of the phonology of the PIE laryngeals.[19] In principle, however, just as the phonology of #HR̥C- need not mirror precisely that of comparable sequences like #HVC- or #CR̥C- (although

[18]Emphasis Lindeman's; similar remarks at 1997:55.

[19]This would not be the only instance in which the observed behavior of laryngeals is poorly understood, as is the case, for example, with the "Saussure[-Hirt] effect" (see Nussbaum 1997), where the laryngeal loss can be described in terms of its phonological environment but has as yet no plausible phonetic explanation, despite various attempted solutions. (An unsatisfactory account is offered by Lindeman [1997:193], discussed below [n. 74]; a different view of the matter has been proposed by J. E. Rasmussen in a number of publications, see e.g. 1997:260f.)

similarities may be expected), it should occasion no surprise if #HR̥C- behaves in some respects like #HRV-. Indeed, previous attempts to formalize the operation of Rix's Law[20] operate with epenthesis as a first step — precisely as in other cases of laryngeal "vocalization" — i.e., the insertion of a new syllabic segment after the laryngeal; this, however, is said to condition in turn the loss of syllabicity of the immediately following resonant.[21] Insofar as epenthesis and syllabicity are involved,[22] it is *a priori* reasonable to assume that the similar behavior of #HR̥C- (or better #HRC-, with Leiden-style "phonemic" notation[23]) vis-à-vis #HRV- is an automatic consequence of phonotactic properties of Greek or PIE: a great deal of recent phonological research on epenthesis, in both prosodic and constraint-based frameworks, has led to just such a conception of the nature of epenthesis processes.[24] In sum, the arguments against Rix's Law based on alleged conflicts with hitherto established "facts" (or *opiniones communes*) about laryngeal

[20]See especially Peters 1980:119ff., building on Joseph 1975.

[21]Viewed somewhat differently, and under the assumption that a sequence like "#HR̥C-" is indeed the correct starting point (more on this below), the process could be described as a switch in syllabicity between the laryngeal and the resonant (i.e. #HR̥C- > #əRC-), comparable to the syllabicity switch in word-final glide-plus-laryngeal sequences in Greek, i.e. **-iH#* > [-i̯ə#] and **-uH#* > [-u̯ə#] (Peters 1980:120; essentially the same syllabicity-switch analysis, including the comparison with final glide/laryngeal sequences, was arrived at, apparently independently, by Meißner [1998a:38]). Lindeman's argument that the different phonetic environments #HR̥C- (with Rix's Law "vocalization" of the laryngeal) and #HRV- (with pre-consonantal laryngeal vocalization) should not produce the same phonological outcome confuses process and result: as already discussed by Peters (loc. cit.), what appears on the surface as laryngeal "vocalization" in these two contexts does not necessarily recover the same underlying phonological process.

[22]See the useful discussion of these factors, in laryngeal contexts, by Hoenigswald (1988).

[23]Cf. Beekes 1994:453, 1995:146 (discussed directly below, **2.4.**).

[24]In a prosodic framework, see e.g. Piggott and Singh 1985, Itô 1989; for a constraint-based approach to Hittite epenthesis and syllable structure, see Kavitskaya 1999; an early "Optimality-Theoretic" study of epenthesis is Piggott 1995, and numerous subsequent studies of epenthesis in that particular constraint-based framework are accessible via the Rutgers Optimality Archive. (I thank Bruce Hayes for helpful discussion of recent phonological approaches to epenthesis.) For general discussion of laryngeal "vocalization" and epenthesis in PIE and Indo-Iranian, see now Kobayashi 2004:132ff.; similarly Lindeman himself (2004), operating with a theory of anaptyxis for certain PIE contexts.

behavior (or based on what are thought to be unlikely phonological developments per se) seem mainly to reflect preconceptions about Greek phonology that may or may not be true. Such arguments can bear little weight in any assessment of the theory.[25]

2.4. *"#h_2R̥C- (> #R̥C-) > Gk. #ἀRC- in any case"*. This claim is expressed most explicitly by Lindeman (1982:60): "since IE. *$r̥$ (and *$l̥$) appear as Gk. αρ, (αλ) in cases where no initial laryngeal is involved, ... there is nothing to prove that a preform *#$r̥ĝ(r)$*ós* (after the loss of the initial **H-* in **H$r̥ĝ(r)$*ós*) could not have yielded Gk. ἀργός. The same objection applies to all of Rix's examples of the types ἀρC-, ἀλC-."[26] This is clearly the weightiest among the counterarguments offered against Rix's Law, if its underlying assumptions are true. The claim that a sequence #h_2R̥C- (the presumed starting point) would simply lose its laryngeal before a syllabic segment (just as in normal prevocalic laryngeal loss) and that the resulting #R̥C- would in any event develop to Gk. #ἀRC- means, in effect, not that the liquid version of Rix's Law is invalid, but that it cannot be proved, thereby draining it of all but a certain theoretical interest.

But the starting point "#h_2R̥C-" itself (or more generally "#HR̥C-") may be illegitimate, as the discussion of epenthesis in **2.3.** has already suggested. Such a notation (which has, to be sure, become standard, apart from "Leiden School" practitioners, from the time of Rix's original treatment) embodies the assumption that a PIE sequence like **h_2RC-* would in the first instance have been automatically syllabified as **h_2R̥C-*, exactly comparable to any sequence **CRC-* (syllabified as **CR̥C-*).[27] But while the non-syllabic character of

[25]An additional phonological counterargument, based on the relative chronology of laryngeal vocalization and the Greek treatment of **R̥*, is discussed in the next section.

[26]Similarly Meid loc. cit. ("Das angenommene Resultat von **h_2R̥*, nämlich *aR*, ist *prima vista* nicht von dem von **h_2eR* oder **R̥* allein zu unterscheiden") and Meißner 1998a:38 ("The majority of Rix's examples have initial **h$_2$-* before **r̥* or **l̥*, and where this laryngeal is not supported by comparative evidence, its existence is not demonstrable as **r̥* and **l̥* generally developed an epenthetic vowel /a/ anyway", and further, in connection with ἀργός and a preform **$h_2r̥ĝ$-*, "the Greek outcome would have been ἀργ- with or without the laryngeal").

[27]Lindeman strongly insists on this point, e.g. at 1994:42f., 1997:55.

the PIE laryngeals is not in doubt,[28] certain of their other properties remain open to discussion. It has in fact been argued that a syllabification of this sort would not have been the immediate result of such a sequence, on the basis of the Rix's Law nasal contexts. These are discussed by Beekes (1994:453, cited above [n. 23]), who points out that a PIE sequence **CnC-* syllabifies as **Cn̥C-* (> Gk. /CaC-/), whereas a Rix's Law sequence like **h₂nC-* cannot be said to syllabify as **h₂n̥C-*: this would predict a Greek result /aC-/, whereas the actual outcome (according to Rix) is /anC-/.[29] This means that (as Beekes writes) "the *n* cannot have been vocalic", and "the actual development *ank-* must be ascribed to a special development ('vocalisation') of the laryngeal". If this is true, it would likewise be improper to assume that the Greek outcome /aRC-/ of a PIE sequence **h₂RC-* results from a syllabification **h₂R̥C-*, rather than from a "special development" related to the properties of laryngeals on the one hand and Greek (or PIE) phonotactic constraints on the other (cf. **2.3.**).[30] This perspective would also provide the solution to an apparent chronological conundrum posed by Meißner (1998a:37): in considering how a laryngeal could have colored the epenthetic vowel that results from *R̥* (in sequences **HR̥-*), he observes: "The problem ... is that the loss of the laryngeals in Greek would have to be incredibly late, posterior to the dissolution of the vocalic resonants". Since there is little support for such a late retention of the laryngeals in Greek, it seems "unwise to claim that laryngeals could have survived until after the dissolution of the syllabic resonants". But, as we have just suggested, Meißner's premise may be false to begin with; the

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[28]As I hope to discuss elsewhere, I am not persuaded by the arguments of Reynolds et al. (2000), whose treatment, in any event, ignores Rix's-Law contexts. See, for the moment, Kobayashi 2004:127ff.

[29]Thus for Rix (1970:90), *ἀμφί* 'on both sides' derives from **h₂m̥bʰí*.

[30]Similarly Beekes 1995:146 on "*HRC-* in anlaut": "Greek vocalized the laryngeal: *e/a/oRC-* (it is therefore incorrect to write *HR̥C-*!)"; cf. Ruijgh (1988:451): "Nous avons écrit *r̥, n̥, m̥* d'après l'habitude, mais il nous paraît plus probable qu'une séquence phonologique comme **h₃rnéw-* se réalisait phonétiquement par une voyelle d'appui entre la laryngale et la liquide ou nasale, non pas par une liquide ou nasale voyelle authentique". Note that examples of the type PIE **HRC-* > Hitt. /ḫaRC-/ do not, *pace* Lindeman (cf. n. 27), demonstrate a PIE syllabification **HR̥C-*; they merely show that the epenthetic vowel that surfaces in this environment in Hittite is not distinguishable from that which arises from sequences like **CRC-* (> **CR̥C-*).

epenthetic vowel resulting from Rix's Law (where the resonant is not necessarily syllabified as *R̥*, despite standard assumptions to this effect) need not be the "same" epenthetic vowel that results from a syllabic resonant in, say, a non-laryngeal sequence like **CR̥C-*.

Although it may at first seem counterintuitive, the position just argued concerning PIE sequences **HRC-* (which need not be assumed to have syllabified as **HR̥C-*) is not in any way weakened — and in a way is strengthened — if Rix's Law does not in fact operate in nasal contexts at all. In an extended and thorough treatment, A. S. Nikolaev (2005) has recently suggested that despite Rix's claims (and, therefore, the arguments of Beekes reviewed above), the Greek evidence for nasals is better analyzed as showing a development **HNC-* > *ἀ*C-. If this is so, it would only confirm the long-standing judgment that, as A. Morpurgo Davies has observed (following C. J. Ruijgh on this point), "the treatment of the nasal and liquid sonants constitute two altogether different problems" (1968:791).[31] There can be little surprise in the eventuality that post-laryngeal **n̥* and **m̥* develop to a single vowel — just as syllabic nasals do virtually everywhere else in Greek (apart from the context **(-)NHC-*; Rix 1992:73) — while post-laryngeal syllabic liquids surface in consonantal form together with an adjacent vowel (cf. **CRC-* > A-I C*a*ρC-/Cρ*a*C-). The ultimate effect of the revision proposed by Nikolaev would thus be to render Rix's Law more natural, by removing its unexpected treatment of syllabic nasals. What remains, then, is the fundamental question of the quality of the vocalic segment that surfaces adjacent to the liquid, in Greek sequences /VRC-/ arising from PIE **HRC-*. If the vowel quality can be shown to be systematically related to the nature of the laryngeal, we are justified in speaking of an epenthetic vowel that is comparable to other instances of laryngeal "vocalization", as opposed to the standard "prop vowel" that might result from a sequence syllabified as **HR̥C-*. In assessing this issue, we will thus restrict ourselves to consideration of the liquid version of Rix's Law.

More specifically, the aim of the discussion that follows will be to evaluate on its own terms — as misguided as they

[31] Similar comments by Peters (1980:121) on chronological and typological differences to be expected in the behavior of syllabic liquids and nasals.

may be phonologically — the particular counterargument to Rix's Law identified at the beginning of this sub-section, i.e. "#h_2R̥C- (> #R̥C-) > Gk. #ἀRC- in any case". We will proceed, then, by granting, for the sake of argument, the initial starting point in a sequence syllabified as #h_2R̥C-. Only in this way will it be possible to compare the "Rix's Law predictions", for the sequences in question, with the "anti-Rix predictions" of those critics of Rix's Law who assume that sequences like **HRC-* must syllabify as **HR̥C-*.

Before proceeding, however, we must first consider more carefully the choice of "#h_2R̥C-" (with second laryngeal) in the phrasing of this objection by critics of Rix's Law. This choice appears to be related to the fact that the cases involving the clearest etymologies (note the use of ἀργός by both Lindeman and Meißner) or (according to Meißner's claim [n. 26 above]) "[t]he majority of Rix's examples" involve precisely this sequence.[32] But in order for this objection to be cogent, it must in principle be stated more generally: according to this claim, then, *any* laryngeal should be lost prevocalically (and hence also before a syllabic resonant), leaving an initial sequence #R̥C- that is presumed to develop to /aRC-/. As it happens, however, the treatment of the initial sequence #R̥C- in Greek is not an entirely straightforward matter.

3.0. It has in fact been assumed by some that the development #R̥C- > #ἀRC- is proper only to Attic-Ionic and West Greek, whereas Aeolic and "Achaean" Greek[33] might be expected to show a development #R̥C- > #ὀRC-. This is evidently the view of those who claim that Hom. ὄρχαμος 'leader, commander' and Myc. *o-ka* '(military) detachment' vel sim. (interpreted as /ork^{h}ā/) show the Aeolic and Achaean treatments, respectively, of a sequence *$r̥g^h$-, with the resulting (Aeolic/Achaean) **ork^{h}ā* (as in Myc. *o-ka*)

[32] Meißner's claim, while arithmetically correct, is tendentiously phrased: Rix adduced a total of seven cases involving **h$_2$r̥*- and **h$_2$l̥*-, as against six for **h$_1$r̥*-, **h$_3$r̥*-, and **h$_3$l̥*-.

[33] I use this designation informally as a convenient cover term that includes Mycenaean, Arcadian, and Cyprian.

comparable to Attic-Ionic ἀρχή 'rule'.[34] If this is so — and the point seems to have gone virtually unnoticed — then there are exactly two sets of crucial test cases for Rix's Law, as follows:

(i) In Aeolic and Achaean, a form like **h₂r̥ĝi-* would be expected to develop, after loss of the initial laryngeal, as /orgi-/, according to the conception of Lindeman and others (**2.4.**), vs. the Rix's Law outcome /argi-/.[35] Nevertheless, and despite well-known problems involving alternations between *o-* and *a*-vocalism resulting from the syllabic liquids in Aeolic and Achaean, all Rix's Law cases with initial sequence #h₂r̥C- develop in these dialects unambiguously and regularly to /aRC-/, not †/oRC-/ (**3.1.-3.4.** below). Similarly, if the Aeolic/Achaean result of initial #r̥C- is #ὀRC-, then Rix's Law cases of the shape #h₁r̥C- should again develop in these dialects to /oRC-/ (and not /eRC-/); yet this does not seem to be the case either (**3.5.** below).[36]

(ii) In Attic-Ionic and West Greek, both **h₁r̥C-* and **h₃r̥C-* should develop to #ἀRC- according to the above assumptions (vs. **h₁r̥C-* > #ἐRC- and **h₃r̥C-* > #ὀRC- by Rix's Law); this prediction is again contradicted by the available data (**3.6.** below).

3.1. The Mycenaean treatment of syllabic liquids is still in some respects controversial, despite a great deal of scholarly attention.[37] The essential difficulty, apart from the possible survival of syllabic liquids as such, is the simultaneous appearance of both *a-* and *o*-vocalism (cf. Meißner, n. 35 above, on Aeolic). As A. Morpurgo Davies observed, in an

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[34]For references to this conception of ὄρχαμος and *o-ka*, see Aura Jorro 1993:19, s.v. *o-ka*. (Different approaches to this material will be referenced below.)

[35]Only Meißner seems to have any inkling of the issue, when he writes that the syllabic liquids generally developed an epenthetic /a/, while admitting that "even in what seems to be genuine Aeolic where the /o/ treatment is more prevalent, /a/ sometimes occurs" (1998a:38, the full extent of his treatment).

[36]The development of #h₃r̥C- is of no consequence, since here the Rix's Law result and the presumed "anti-Rix" result (according to this version of the development of initial #r̥C- in Aeolic and Achaean) would coincide, as /oRC-/ in both cases.

[37]See Bartoněk 2003:135, with references (to which add Thompson 1996/97, Kazanskij 2001:86f., 93ff.).

important discussion: "...we must reckon with different vocalisms as the result of originally unitary phonemes: we have ... *qe-to-ro-po-pi,* "quadrupeds" with *k^{w}etro-* < **k^{w}et(w)r̥-* vs. *po-da-ko,* i.e. *podargos* < **pod(h_2)r̥gos*" (1985:80). Yet the *a*-vocalism in the second member of the boonym *po-da-ko* /Podargos/ 'Whitefoot'[38] appears in initial position of that morpheme, whether this is conceived of as based on the precursor of ἀργός or ἀργι- (for the latter, cf. Hom. ἀργίπους 'swift-footed' and the *ἀργίπεζα that underlies ἀργυρόπεζα 'id.'[39]); and if this presumed /arg-/ form indeed continues a zero grade,[40] it could simply reflect a regular development according to Rix's Law. Note further that in J. L. García Ramón's thorough (and on the whole highly convincing) study of the development of the syllabic liquids in Mycenaean (1985), the sole group of forms on which he hesitates to pass judgment consists of those with possible initial sequence #r̥C-: in fact, *all such forms show* a-*vocalism,* which García Ramón thinks "may be due to an initial

[38]Not 'schnellfüßig' (Bartoněk 2003:408, 413), despite Hom. Πόδαργος (horse name); cf. πύγαργος [Hdt.+] 'white-rump' (name of an antelope and various birds). With a similar structure also Myc. *to-ma-ko* /Stomargos/ 'Whitemuzzle' (cf., with a different semantic development, στόμαργος [Aesch.+] 'prating'), although the interpretation of *tu-ma-ko* is less clear (Aura Jorro 1993:375, s.v.); see Killen 1992/93 on the semantics of Mycenaean boonyms.

[39]For *ἀργίπεζα see Lamberterie 1978:257 and Peters 1980:227, following Bader 1971:206f.

[40]Cf. n. 17 above. The derivational history of Myc. *po-da-ko* and *to-ma-ko* remains, moreover, somewhat unclear. Meißner and Tribulato (2002:297, 315) are no doubt correct in interpreting these forms, at some level, as endocentric "adjectival" determinative compounds based on an original syntagm with accusative of respect: roughly πόδ' ἀργός πόδαργος. One might, however, expect to have seen the adjective as the first member (cf. the closely related bahuvrihi type Myc. *re-u-ko-nu-ka* 'with white *onukhes* [tassels vel sim.]', Hom. λευκώλενος 'white-armed', etc.), except that, as observed by Hajnal (1995:137n173), Mycenaean possessive compounds with thematic first member seem to be restricted to second members with initial /o-/ (as in *re-u-ko-nu-ka* above: /leuko-/ + /onukh-/), thus excluding e.g. †*a-ko-po* /argopōs/ or the like. Likewise conspicuous by their absence are possessive compounds with "Caland stem" adjectival first member, i.e. precisely of the type ἀργίπους — thus it seems difficult to exclude the possibility that a form like *po-da-ko* might have had a first member **argi-* (with zero grade source, vs. **argo-,* ambiguous between full grade and zero grade) somewhere in its derivational background. (On the close relationship between endocentric determinative compounds and bahuvrihis, see Meißner and Tribulato, locc. cit.)

laryngeal".[41] As for *po-da-ko,* García Ramón observes that this is a prime instance of a form in which "the possibility of initial *$*h_2r̥$-* (or *$*h_2er$-*) can not be ruled out" (1985:211n57), in view of the initial laryngeal confirmed by Hitt. *ḫarki-* (cf. **2.1.**). And while the thrust of his discussion primarily addresses zero-grade forms, García Ramón properly leaves open the possibility of full-grade *$*h_2erC$-* for some of the forms in question, as indeed for *po-da-ko*. We have in fact seen (n. 17 above) that, despite the readiness of Lindeman and other critics of Rix's Law to embrace Gk. ἀργός (and by extension, one assumes, its Mycenaean antecedent) as a bona fide zero grade, this interpretation is open to some question, although ἀργι- (and the **argi-* that may underlie the second member of Myc. *po-da-ko*) would meet no such objection.

3.2. This is not, however, the only instance of a Mycenaean (and Arcado-Cyprian) /aRC-/ form that could in principle derive from a zero-grade *$*h_2R̥C$-* via Rix's Law, and which fails to show a development to /oRC-/, as per the anti-Rix counterclaim (according to the version under discussion of how #R̥C- develops in Greek). For *$*h_2l̥k$-*,[42] note Myc. *a-ki-to* (dat. sg.) /Alkit^{h}ōi/ (cf. post-Myc. ᾽αλκίθοος), as well as Arc. ᾽Αλκι-[43] and ᾽Αλαλϙο[[44] (not †᾽Ολκι-, †᾽Ολολϙο[). Even if some forms of this root might show influence from a full-grade *$*h_2elk$-*,[45] it is not credible to interpret a first compound member /Alki-/ in this way. As for Meißner's skepticism about the existence of initial *$*h_2$-* in this root (noted above [**2.1.** with n. 13]): his appeal to the absence of long augment in Ved. *árakṣa-* is not decisive, as Rix had already pointed out;[46]

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[41] García Ramón 1985:216, similarly 211f. with n. 57; his discussion makes no mention of Rix's Law as such.

[42] See Rix 1970:87f. on ἀλκί, ἀλκή 'bulwark; strength', and cf. *$*h_2elk$-* 'abwehren' (LIV 264).

[43] ᾽Αλκιππος, ᾽Αλκιστρατος, ᾽Αλκιας, etc.; cf. also Arc. ᾽Αλκμ!ν, with a formation pointing to a zero grade abstract *ἀλκμ% (< *$*h_2l̥k$-méh_2*) + -ϝvn, of the type Μαχ%ϝων (see Dubois 1986:1.37).

[44] Mantinea, 6th c. (Dubois 1986:2.91f.).

[45] See Leukart (1994:227) on *a-ka-to-wa* /Alkat^{h}oṵās/ (cf. post-Myc. ᾽Αλκάθοος); /Alka-/ perhaps related to the strong form of a root noun, e.g. acc. sg. *$*h_2élk$-m̥?* But the form is ambiguous in various ways, see Meißner and Tribulato 2002:303.

[46] "in ai. *árakṣa-* (Ipf.) und *rarakṣ-* (Perf.) mit kurzem *a* vor *r* ist freilich keine Laryngalwirkung (mehr?) zu spuren" (1970:87).

and indeed, the long-vowel reduplication in the perf. mid. participle *rārakṣāṇá-* (not mentioned by Meißner) could well reflect a laryngeal lengthening (on which see Kümmel 2000a:413). Moreover, the Sanskrit verb *rákṣati* 'protects' (together with its Greek cognate ἀλέξω 'ward off') in and of itself serves to undermine Meißner's objection: if the root of the Gk. ἀλκ- forms is simply **alk-* (as Meißner seems to be claiming), this effectively removes the possibility of comparing Gk. ἀλκ- with its evident congeners ἀλέξω and *rákṣati,* since alternations **alk- : *alek-s-, *alk- : *lek-s-, *al-k- : *al-ek-s-* (etc.) are anomalous and indefensible, as opposed to the well-supported "Schwebeablaut" pattern **h₂elk- : *h₂lek-s-* with *s*-enlargement (Schindler 1970/72:152), cf. **h₂eu̯g- : *h₂u̯eg-s-* 'increase' (Lat. *augeō,* Gk. αὐξάνω *:* ἀ(ϝ)έξω etc.). Nor would a **h₁alk-* be an improvement, as its enlarged stem **h₁lek-s-* cannot account for the ἀ- of ἀλέξω (at least in a "triple representation" model of Greek laryngeal vocalization).

3.3. Three examples based on material not found in Rix's original treatment can be mentioned very briefly, given indeterminacies in the make-up of the first two, and the relative clarity of the third:

(i) If one accepts Klingenschmitt's attractive derivation of Gk. ἄρχω 'begin, rule' from **h₂r̥-skʰé/ó-,*[47] the regular appearance of /arkʰ-/ (not †/orkʰ-/) in Achaean, where this form is abundantly attested, would again invalidate the anti-Rix claim.[48]

(ii) Att. ἀρά beside Ion. ἠρή 'prayer, curse' point to a Pr.-Gk. **aru̯ā́,* the accent of which suggests a zero grade **h₂r̥-u̯éh₂;*[49] the initial laryngeal is guaranteed by CLuv. *ḫīrūn/t-*

[47]To the root **h₂er-* of ἄριστος 'best, first (in rank)'; 1974:274n1. (This analysis implies that the conception of Myc. *o-ka* noted above [**3.0.**] is not in fact correct; more on this point below, in **4**.)

[48]Myc. *a-ke-wa-to* /Arkʰeu̯astos/ (García Ramón 1990/91), *a-da-ra-ko* [/Andrarkʰōi/, *ta-ta-ke-u* /Stātarkʰeus/ (cf. post-Myc. Στήσαρχος), *qi-si-ja-ko* /Kʷĭsiarkʰos/ (cf. post-Myc. Τείσαρχος, Τιμήσαρχος and Cypr. *ti-ma-ra-ko-se* below; Vegas Sansalvador 1999), etc.; Arc. Ἀρχιδαμος, Ἀρχιππος, Ἀρχιφαης, etc. (cf. also dat. pl. ἀρχουσι [Dubois 1986:2.167f.]; Cypr. *a-ra-ko-se* /arkʰos/ (Masson 1986:455f.), names in *a-ra-ki-* /Arkʰi-/, *a-ra-ko-* /Arkʰo-/, *ti-ma-ra-ko-se* /Ti1markʰos/, etc.

[49]Zero grade reconstructed also by Oettinger (1979:345).

'oath'.[50] The Achaean treatment is shown by Arc. καταρϝος 'cursed', Cypr. *a-ra-wa-sa-tu*[51] and *a-ra-wa-to,*[52] and perhaps also Myc. *ka-ta-wo* and/or *ka-ta-wa.*[53] But a full-grade *$*h_2er$-u̯$é$h_2$* is possible, and the etymology itself has been dismissed by Lindeman (1997:82 with n. 80).[54]

(iii) Gk. ἀρνέομαι 'deny' has been plausibly taken to reflect a thematized nasal present stem *$*h_2r$-né-s-* (LIV 284 s.v. *$*h_2res$-*, with further references); the laryngeal is supported by the vowel length in the reduplicating syllable of GAv. *rārəšiieiṇtī* 'werden abtrünnig' (see LIV ad loc., n. 2, for details). For the Achaean treatment, note Arc. ἀρνε̄[τοι] (not †ὀρν-), in fragments from an archaic *lex sacra* (Schwyzer DGE 661g.20, Mantinea, 6th c.; see Dubois 1986:2.114 for the reading and interpretation).

3.4. Little information from Aeolic is available for the material in **3.1.-3.3.**, and as usual, literary Aeolic must be evaluated bearing in mind the possibility of epic influence. This consideration, however, can be exaggerated: thus, despite the predictable similarity between [κλ]ῦθί μ' ἄρας (Sa. 86.5 Voigt) and various Homeric passages (Voigt cites θεὰ δέ οἱ ἔκλυεν ἀρῆς, δ 767), there is no reason to believe that the form of Lesb. *ἄρᾱ* (not †*ὄρᾱ*, cf. **3.3.**(ii)) has been "Ionicized". Other Aeolic data are entirely consistent with the picture sketched above: thus forms of ἄρχω (never †ὄρχω, cf. **3.3.**(i)) are frequent in both literary and epigraphic Aeolic (similarly ἀρχι-, ἄρχων, ἀπάρχων in inscriptions);[55] cf. also Thessal.

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[50]See Watkins 1993:469f. for the Luvian etymology, and Rix 1993:333 on the Greek forms.

[51]3 sg. aor. mid. /ārṳasatu/ 'hat gebetet, ein Gelübde abgelegt' vel sim. (Egetmeyer 1992:17, s.v.).

[52]MN /Arṳatō/ (gen. sg.).

[53]See e.g. Chantraine, DELG s.v. ἀρά (*ka-ta-wo* perhaps /Kataruos/, cf. Arc. καταρϝος) and Aura Jorro 1985:330 s.vv. *ka-ta-wa* (perhaps /Katarṳās/) and *ka-ta-wo* (/Kataruos/ among various other possibilities).

[54]Lindeman does not pronounce on his preferred etymological connections for ἀρά. Note further that Watkins' analysis of the Luvian form depends on Eichner's Law — another proposal regarding laryngeal behavior that Lindeman categorically rejects, as I discuss elsewhere ([forthcoming]).

[55]Note Hamm 1957:28, who lists ἀρχάω[(Alc. 67.5 Voigt/Campbell, though this may well be Ἀρχάω[, see Voigt and Campbell ad loc.) in her discussion of

Ἀργεια (Schwyzer DGE 583; Larisa, 5th c.) (not †Ὀργ-, cf. **3.1.**), Boeot. Ἀμφαλκεῖς (DGE 455; Tanagra, 5th c.), Δαμαλκος (DGE 478B; Thespia, 5th c.) (not †-ολκ-, cf. **3.2.**).

An item that can conveniently be added here is the word for 'bear': here it is noteworthy that Lindeman (e.g. 1997:54) operates with a reconstruction **Hr̥tḱo-* that is virtually identical to that proposed by Rix ("**h₂r̥ḱpos* [< **h₂r̥tḱos*?]", 1970:85), apart from the specification of laryngeal color. Once again, if initial **r̥-* produces *o*-vocalism in Aeolic and Achaean, critics of Rix's Law who believe that "#h_2RC- (> #RC-) > Gk. #ἀRC- in any case" would need to account for why this form appears pandialectally as ἄρκτος.

3.5. For the initial sequence #h_1RC-: if Rix's etymology of ἔρχομαι is correct, this form's *e*-vocalism in Lesbian (Sa., Alc.) and Achaean (cf. the Myc. toponym *e-ko-me-no* /Erk^{h}omenos/ and MN *e-ko-me-na-ta-o* /Erk^{h}omenātāo/)[56] runs counter to the "anti-Rix" expectation, assuming the treatment of initial **r̥-* in Aeolic/Achaean to be /or-/. But the etymology is uncertain (**2.1.**), so we need not insist on this item.

There are few data for this sequence otherwise, and we reserve an Attic-Ionic case and a possible Mycenaean case for discussion below (**3.6.**(i), **6.3.-6.4.**). The best example incorporating material outside Attic-Ionic is the word (or rather words) for 'male' (adj.). But the profile of this material is complex, and calls for a somewhat extended discussion. The relevant background information can be sketched as follows.

Lexica typically organize the Greek words for 'male' around the epic and Attic forms: so e.g. Frisk, GEW I.152 s.v. (Hom.) ἄρσην, listing in addition "ἄρρην (att.), ἔρσην[57] (ion. lesb. kret. usw.), ἄρσης (lak.)", and further a form with comparative suffix, as in (to continue with Frisk's notations) "ἀρρέντερος 'männlich' (ark.), ἐρσεναίτερος (el.)". (More on the Arcadian form below.) The etymon of this material is said to be the word for 'man' that appears in IIr. forms like OP *aršăn-*

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the *o*- vs. *a*-treatment of syllabic resonants — the sole instance she provides for word-initial position; cf. also ἀρξαν[, Alc. 113.1 Voigt/Cambell.

[56]Mycenaean already shows the secondarily assimilated form /Ork^h-/ (MN *o-ko-me-ne-u* / Ork^{h}omeneus/), as in the post-Myc. place name Ὀρχόμενος beside Ἐρχόμενος (Arcadia, Boeotia).

[57]Accent sic; further below on the question of the accentuation.

/YAv. *aršān-* 'männlich, Mann', Ved. *ṛṣabhá-* 'bull', Arm. *aṙn* (gen.-dat.-loc. sg. *aṙin*) 'ram', from a PIE hysterokinetic **h$_1$ṛs-én* 'männliches Tier' (*ṛṣabhá-* < **h$_1$ṛs-n̥-b^{h}ó-*).[58] Finally, handbooks often note (to quote Frisk once more) "die synonyme Reimbildung aind. *vṛ́ṣaṇ-*",[59] and point to a possible connection between ἄρσην etc. and the words ἀρνειός [Hom.+] 'ram' and possibly ἀρνευτήρ [Hom.+] 'acrobat, diver',[60] although those words have also been compared with ἀρήν (Cret. ϝαρην) 'sheep, lamb' (Ved. *úrā* etc., < **u̯r̥h$_1$-én*; Balles 2002:3).

In a dense and tightly-argued study — to which the reader is referred for many additional details — M. Peters (1993) sought to show that the Greek words for 'male' should actually be separated into two original etyma, derived from the PIE rhyming synonyms noted above, and distributed as follows:

(i) Hom. ἄρσην/Att. ἄρρην (and other forms with *a*-vocalism attested in relatively late inscriptions in Doric koiné: e.g. Lac. ἀρσης, cited above) < **h$_2$u̯r̥s-én* (thus with Ved. *vṛ́ṣaṇ-* etc.);[61]

(ii) the forms with *e*-vocalism (Lesb. ἔρσην etc.) < **h$_1$ṛs-én* (thus with YAv. *aršān-* etc.), which thereby provide an additional example of Rix's Law, of precisely the sort under discussion (cf. **3.0.**(i), second part).

Several additional details may be noted:

(a) The alleged Arcadian form "ἀρρέντερος" (cf. Frisk, above) does not exist as such: in the inscription in question,[62] the word is attested (lines 21 and 27) as part of a sequence τορρεντερον, traditionally assumed to reflect a crasis product (by contraction) of τὸ ἀρρº. As Peters argues, however (1993:380 with 379n30), it is far more likely that this sequence stands for

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[58] On this and other hysterokinetic *n*-stems designating "Tiernamen", of which at least a half dozen are reconstructable for PIE, see recently Balles 2002:2f.

[59] I.e. 'bull', < hysterokinetic **h$_2$u̯r̥s-én* 'männliches Tier" (Lat. *uerrēs* 'ram' etc., cf. Balles 2002:2).

[60] In part used for animal names (esp. birds, fish), cf. the later-attested verb ἀρνεύω 'tumble, dive' (Lycophron) that may originally have meant 'leap (like a ram)'.

[61] I.e. ἀρσ-/ἀρρ- < Pr.-Gk. **u̯r̥s-*; the absence of "prothetic vowel" (< **h$_2$-*), according to Peters, is regular in this context (1980:23n18). On the secondary *-ς* of Lac. ἀρσης: Peters 1993:378n26.

[62] IG V^2 262 = Dubois 1986:2.94ff., Buck *Dialects* 17, Schwyzer DGE 661 (Mantinea, 5th c.).

τ' ὀρρº (via elision). If so, the absence of digamma would appear to constitute a problem, though perhaps not a serious one. Initial digamma is otherwise regular in this text; but this is also the inscription with the notorious form <ϝοφλεκοσι>, with etymologically incorrect (and probably hypercorrect) digamma. Given other evidence for precocious loss of digamma before /o/, Peters suggests that Arc. *ὀρρ- here can indeed derive from an earlier *u̯ors- (< *[h₂]u̯r̥s-), and can thus belong with the material in (i) above. Meißner suggests (1998a:39 with n. 12) that τορρεντερον may instead be interpreted as standing for τὸ ἐρρº, by contraction — as, in fact, Peters had already considered as a technical possibility (1993:376n13). As this would eliminate the problem with digamma and place the Arcadian form with (ii) above, it poses no particular difficulty. Still another possibility (Peters 1993:380) would involve the medial contraction result of a sequence /tou̯arr-/, /tou̯err-/, or /tou̯orr-/ in a lexicalized "phonological word", with the (proclitic) article and the following word treated as a fixed syntagm; this would again remove the problem with the digamma, as either /u̯arr-/ or /u̯orr-/ permits easy alignment with the material in (i). Thus, while the precise interpretation of this Arcadian datum remains uncertain, its form (whether *(ϝ)αρρεν-, *(ϝ)ορρεν-, or *ἐρρεν-) does nothing to weaken Peters' analysis of the material in (ii) as a further instance of Rix's Law.

(b) The dossier of forms related to this problem has now been enriched by the new Thessalian form ὀρσεν (neut.), from a 3rd-century stele (GHW 5765, Larisa).[63] The context is clear (lines 1-2): a male animal (ὀρσεν) is offered to Zeus, and a female (θειλυ) to En(n)odia. As discussed by García Ramón ([forthcoming]:§9), the *o*-vocalism here can be considered to reflect the regular Thessalian result for *-r̥-, at least in the environment of a labial; thus the form can best be interpreted as reflecting a /(u̯)orsen/ < *u̯r̥s-en, comparable to Hom. ἄρσην etc. as in (i) above, under Peters' analysis of that material. As

[63]I am deeply grateful to José Luis García Ramón for generously providing me with a preliminary version of his forthcoming paper on Thessalian, containing a discussion of this text (as read by both him and by Bruno Helly, to whom I also extend my gratitude), and for other assistance connected with the material in this section.

for the lack of digamma: as García Ramón observes, this is entirely unproblematic in this 3rd-century text, as digamma was maintained consistently in Thessalian only until the 4th century. (On digamma, see also (d) below.)

(c) The handbook accentuation ἔρσην (as appears, for example, in the Ionic of Herodotus, cf. Frisk, above; apparently also assumed by Meißner) gives an incomplete and perhaps wholly inaccurate picture of the accentual facts about the *e*-grade forms (apart from Lesb. ἔρσην, where the retraction is regular). As discussed in great detail by Peters (1993:375ff. and passim), following earlier analyses by Solmsen, there is extensive dialectal evidence for oxytone accentuation, given the direct testimony of forms like Laconian / Messenian ἰρήν 'full-grown youth' (< Pr.-Gk. **erzḗn*)[64] and the forms pointing to a Pr.-Gk. **erzaբʰo-*[65] that is generally agreed to be comparable to Ved. *ṛṣabhá-* (cf. above). The treatment of PIE medial **-rs-* in these forms points directly to oxytone accent (see Peters 1993:376n13, with references), while a form like ἔρσην (Hdt.) would be based on a contamination-product involving the preform of Ion. ἄρσην.[66] Thus Lindeman's characterization of Gk. (dial.) "*ersēn*" [sic] as a form "of uncertain accentuation" (1997:56) is a potentially deceptive oversimplification. The presence of direct evidence for oxytone accent, in turn, provides part of Peters' argument — but only a part of it — in favor of original hysterokinetic inflection for the *e*-grade forms (as in (ii) above); see Peters 1993:392.[67]

(d) If Peters' analysis of the material in (i) is correct, one might well expect to find traces of the Pr.-Gk. initial **u̯-*, in particular through Homeric scansions favorable to the

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[64]See Frisk, GEW I.466 s.v. εἰρήν.

[65]Ion. Εἰραφιώτης [hHom.+], Lesb. Ἔρραφ^{o} (epithet of Dionysus; Alc. 349c Voigt/381 Campbell), Ion. Εἰραφιων (month name, Amorgos).

[66]The situation is in fact somewhat more complex: see Peters 1993:374n5, 398 on the probability that the original **h₁ŕs-n̥-bʰó-* 'bull' underlying the forms in n. 65 underwent an analogical accent shift to **h₁ŕs-n̥-bʰo-* (after **h₁élh₂-bʰo-* 'deer' [ἔλαφος], **h₁éri-bʰo-* 'kid' [ἔριφος]), whence Pr.-Gk. **érsabʰo-*, later remodeled to **érzabʰo-* after **erzḗn*.

[67]As for Meißner's suggestion that ἄρσην beside ἔρσην [sic] could simply result from different levelings of an original proterokinetic paradigm: "eine proterokinetische Flexion kommt bei einem geschlechtigen *n*-Stamm ... kaum ernsthaft in Betracht" (Peters 1993:392n78).

presence of digamma. Peters provides four types of responses to this expectation; briefly:

α. It is easy to imagine that by the time of late Proto-Greek, inherited **u̯r̥sēn* (or, in Peters' notation showing the pre-Greek result of the earlier syllabic liquid, **u̯ársēn*) could have lost its initial glide by contamination with **erzḗn* (1993:378).

β. The Attic-Ionic and Doric koiné inscriptional evidence for ἄρσην/ἄρρην and similar forms cannot by its nature provide testimony for or against digamma, while the Arcadian evidence (see (a) above) can be interpreted as favorable to digamma (1993:378-381). To this can now be added the evidence of Thessal. ὀρσεν ((b) above).

γ. If the contamination in *α*. (above) did not take place: digamma is actually excluded for only three of the 11 instances of Hom. ἀρσεν- (none formulaic and all in the *Odyssey*), given certain considerations (again following earlier work by Solmsen) having to do with pre-Homeric syllabification patterns (1993:381-386).

δ. The forms ἀρνειός and ἀρνευτήρ (see above) also belong with the material in (i); lack of digamma in Homer can be due to dissimilatory loss (pre-Hom. **u̯arnĕu̯-* > **arnĕu̯-*), just as occurred in the word for 'wool' (and several other forms with similar structure): **u̯eru̯-* (cf. Myc. adj. *we-we-e-a*) > (Hom., Aeol, Cret.) **eru̯-*. Moreover, the antecedent of pre-Hom. **u̯arnĕu̯-* is directly attested in Mycenaean (*wo-ne-we* 'young rams'). (Peters 1993:386-392)

Lindeman (1997:56) dismisses Peters' metrical arguments (*γ*. above), but fails to comment on other evidence, such as that provided by ἀρνειός/ἀρνευτήρ and the direct testimony of Mycenaean (*δ*.). As for the *e*-grade forms themselves, which constitute the crux of the matter: Lindeman's own explanation (1997:56f.) is entirely unacceptable. It is sufficient to note that his rather complex "chain of phonologic events" (Lindeman's term) ultimately depends, first of all, on the improbable claim that "Gk. **ersap^{h}o-*" (Lindeman's notation, cf. (c) above), which derives from an inherited zero grade that in Lindeman's system would have ended up as **arsap^{h}o-*, acquired its *e*-vocalism (and not just its accent, cf. n. 66) as the result of analogical influence from **h$_1$élŋ̍-b^{h}o-* and **h$_1$éri-b^{h}o-*; and secondly, his scenario depends

on the still more improbable claim that this marginal term is the motivating force for the analogical introduction of *e*-grade into some forms for 'male', most of which regularly developed to *arsen-* (< **h₁r̥s-en-*),[68] apart from the *e*-grade "variant form", which Lindeman characterizes (oddly, in view of the rather widespread attestation patterns studied by Peters) as "of limited distribution". This distribution prominently includes West Greek (see below) — where no trace of "Gk. **ersapʰo-*" is in fact attested. No credence can be placed in such an account of this material.

In conclusion, then, Peters' analysis of the *e*-grade forms for 'male' provides a well-supported instance of a form in #h₁R̥C- that, counter to the "anti-Rix" assumptions of the version of the development of **r̥-* in Greek under consideration, develops neither *o*-vocalism in Aeolic (cf. Lesb. ἔρσην)[69] nor *a*-vocalism in Attic-Ionic[70] and West Greek (cf. Lacon. / Messen. *ἰρήν*, and forms with *e*-vocalism attested in Cretan, Cyrenaean, Coan, Elean, etc.; Peters 1993:376n13).

3.6. As noted above (**3.0.**(ii)), the other set of crucial cases, for this version of the development of initial **r̥-* in Greek, comes from the treatment of **h₁R̥C-* and **h₃R̥C-* forms in Attic-Ionic and West Greek, where the outcome #ἀRC- for both would be predicted by critics of Rix's Law, as opposed to Rix's claimed outcomes ἐRC- and ὀRC-, respectively. Apart from his discussion of the word(s) for 'male', Meißner addresses such forms only by attempting to call into question the existence of two separate roots **h₁er-* and **h₃er-* (one of the most prominent proposals of Rix's original treatment), a topic that we will consider in section **6.** below. For material that does not depend on those two roots, however, the anti-Rix prediction — according to this version of the treatment of initial **r̥-* in Greek — is not borne out. Here it should suffice

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[68]No explanation is offered for the unexpected accentuation of ἄρσην/ἄρρην, if — as Lindeman seems to agree in the end — the form is based on an original hysterokinetic paradigm.

[69]The Arcadian form is at worst unclear, and may even reflect *e*-vocalism ((a) above), and Thessal. ὀρσεν ((b) above) belongs with the other etymon for 'male'.

[70]Hom. ἄρσην/Att. ἄρρην belong to a different etymon, and cf. Hdt. ἔρσην (unless this is due to a Doric substrate, cf. Peters 1993:378n25).

to recall several of Rix's own examples (one of them with an important update):

(i) An isolated and attractive **h₁R̥C-* > ἐRC- case (Rix 1970:97f.) appears in Att. ἐρέσθαι/Ion. εἰρέσθαι 'inquire' (aor.) beside Hom. (pres.) ἐρέω, Cret. ἐρυταί 'ζητηταί', Att. ἐρευνάω 'search for'— forms for which the best "face-value" analysis points to a thematic aorist stem **h₁ru̯-é/ó-* beside a thematic present **h₁réu̯-(e/o-)* (most likely thematized from an earlier athematic present, cf. subjunct. ἐρείομεν [A 62] < *ἐρέϝ-o-μεν, with metrical lengthening).[71] To my knowledge, Rix's rigorous morphological arguments in favor of analyzing Gk. *ἐρϝ- beside *ἐρεϝ- as reflexes of zero grade and full grade forms, respectively, of a root **h₁reu̯-* have nowhere been countered by critics of Rix's Law. Rix has also anticipated an analogical counterargument, according to which an original aor. *ἀρϝέ/ό- would have been remodeled to *ἐρϝέ/ό-, after the present stem *ἐρεϝ-(ε/o-): "bei einer Analogiewirkung (für *ἀρϝ-, wenn dies als lautgesetzlich anzunehmen wäre) hätte man die Übertragung des ganzen *ἐρεϝ- erwartet". Note also that such an analogical scenario would be further weakened by the necessity to claim that the very same remodeling (yet again without the "Übertragung des ganzen *ἐρεϝ-") took place in the related verb Att. ἐρωτάω/Ion. ερ ωτάω 'ask', which — despite the long-standing difficulty in explaining the obscure stem-final -ω- — must somehow also be based on a zero-grade stem **h₁ru̯-é/ó-* at least formally comparable to that seen in Att. ἐρέσθαι/Ion. ερ έσθαι.[72]

(ii) According to Rix (1970:93), an isolated **h₃R̥C-* > ὀRC- case appears in (gen. sg.) ὀργυιᾶς (nom. ὄργυια 'arm-span;

[71] Similarly Ruijgh 1988:451.

[72] This material, to be sure, is handled in an entirely different way by LIV (251, s.v. 1. **h₁reh₁-*); on ἐρωτάω, see also Rasmussen 1986, 1990:447 (who nevertheless accepts Rix's analysis of εἰρέσθαι and ἐρέω). In my view, however, both the LIV treatment and Rasmussen's analysis of ἐρωτάω are highly problematic, as I discuss in detail elsewhere (2002[2004]). It is sufficient to note here that the LIV approach via a root **h₁reh₁-* (an idea credited to E. Tichy) requires separating Gk. εἰρέσθαι/ἐρέω and ἐρωτάω from the diphthongal forms in Greek (e.g. ἐρευνάω, Cret. ἐρυταί) and elsewhere (e.g. ON *rauna* 'investigation') that are so attractively integrated by Rix's reconstruction.

fathom' [Hom.+], cf. ὀρέγω 'extend') < **h₃ŗ̥ĝ-us-i̯éh₂-s*. While the analysis of ὀργ- (< **h₃ŗ̥ĝ-*) as a zero-grade pendant to ὀρεγ- (< **h₃reĝ-*) is immediately appealing, the morphological structure of such a "**h₃r(e)ĝ-u(-?)s-i(e)h₂-*", on which Rix does not comment, is decidedly obscure. As noted by Nussbaum (1986:147), the form would seem to involve a rather complex derivational history, with an *s*-suffixed *u*-form of some sort further extended by the "devī" suffix.[73] But a manifestly superior analysis has been proposed by C. de Lamberterie (1991/93:129), who argued that ὄργυια is a substantivized feminine dual to an original *u*-stem adjective *ὀργύς (< **h₃ŗ̥ĝ-ú-*, cf. Ved. *ŗjú-* = Av. *ərəzu-* 'straight'), ultimately based on a syntagm *ὀργυιὰ χείρ 'bras tendu' (cf. Ved. *ŗju-hásta-* 'qui a le bras tendu'). In this way, the zero grade finds attractive morphological justification, as well as extra-Greek support. With this adjustment to Rix's original proposal, then, ὄργυια emerges as one of the strongest examples of Rix's Law.

It is telling, in fact, that Lindeman goes so far as to admit (1997:128) that Lamberterie's analysis is "possible"; but he then dismisses it without argumentation, on the *a priori* grounds that if Rix's Law is "phonologically unjustifiable" (as he claims), then Lamberterie's analysis is "equally unfounded". Lindeman's own account of ὄργυια (1990, repeated at 1997:127) is primarily designed to justify the initial *o*-vocalism of ὀρέγω < **h₃régĝ-e/o-*, which in Lindeman's monochromatic version of Greek (cf. n. 8 above) should have developed to †ἀρέγω. His proposal, however, is wholly incredible, as it depends, first of all, on the traditional view of ὄργυια as an original perfect participle, together with a complex sequence of analogical developments that crucially depend on an unattested perfect stem **orōrog-*, the alleged source of the **orōreg-* in Hom. ὀρωρέχαται (Π 834), ὀρωρέχατο (Λ 26). In all of this, Lindeman fails to address the objection already raised by Lamberterie (1991/93:129n4), to the effect that the root **h₃reĝ-* "ne semble pas avoir eu de parfait à date ancienne" (so also LIV 304, s.v.), and that the Homeric perfect "se dénonce par son vocalisme comme une formation de date grecque bâtie sur ὀρέγω". Lindeman's account further depends on an

[73]For the more traditional account in terms of an original perfect participle, see further below.

improbable and unsupported phonological claim (the same one that is said to account for the "Saussure-[Hirt]" pattern of laryngeal loss, cf. n. 19 above).[74] There is thus little reason to prefer Lindeman's method of explaining ὄργυια over Lamberterie's elegant and morphologically transparent analysis, which yields a particularly convincing Rix's Law example.

(iii) It is axiomatic that the initial syllable of athematic nasal presents is based on a zero-grade stem; there can thus be little doubt as to the zero grade of the nasal present that

[74]Lindeman teaches that the late IE sequence **o̯rĝ-* (sic, with hiatus) resulting from the loss of the prevocalic (or rather "presyllabic") laryngeal in an earlier sequence **h₃e-h₃r̥ĝ-* would in turn develop to pre-Greek **org-*, "by removal of the hiatus [o̯r̥]", with reference to his discussion of laryngeal hiatus forms (1997:58ff.). (Similarly */tórHmo-/ > *[tórHm̥o-]* ("according to Sievers' Law") > **tormo-* and the like [1997:193], cf. n. 19 above.) So far as I can see, however, Lindeman's survey of the removal of laryngeal hiatus adduces no Greek form with a comparable sequence. The closest comparandum he cites appears to be **h₂u̯eh₁n̥to-* 'wind' (with no Greek reflex), discussed in another context (1997:123), where again it is claimed that "the hiatus **-en̥-* has been removed ... in the preform of Lat. *uentus,* Goth. *winds,* etc., i.e. **wen̥to-* > **wen to-*". (In his earlier treatment of this material [1990:168], Lindeman had listed "Gk. ptc. ἀ_ϝεντ-" as a comparison to Lat. *uentus* < **wen̥to-* [with hiatus]. This was entirely improper, as the synchronic pres. participle of ἄ(ϝ)ημι tells us nothing about the Greek development of PIE **h₂u̯eh₁n̥to-* 'wind'. The comparison does not reappear in his 1997 treatment.) These forms, of course, can be explained in an entirely different way, according to which the syllabic resonant following the laryngeal develops in its usual post-consonantal fashion, whence e.g. pre-Lat. **u̯e.ento-* (with hiatus) > **u̯ēnto-* > Lat. *uentus* (with shortening by Osthoff's Law; see e.g. Meiser 1998:75, 110). For Greek, the treatment of post-laryngeal R̥ is the same as post-consonantal R̥ at least in word-final position, as is shown by a form like κύαρ 'hole, eye of a needle' (cf. Av. *sūra-* 'hole', Arm. *sor* 'id.' etc., from a root **ḱeu̯H-*; Nussbaum 1997:183), the simplest analysis of which would operate with an *r*-stem (or *r/n*-stem) source **ḱúH-r̥*. But in medial position, the word for 'month' shows a different treatment, and one which falsifies Lindeman's claim: for a PIE sequence **meh₁-n-s-* (comparable to that in 'wind' above), Lindeman's assumptions would predict, first, a syllabification **meh₁-n̥-s-*, whence a hiatus-form **men̥s-* (which is indeed the precursor of the Indo-Iranian form **maäs,* as shown by disyllabic scansion of GAv. *må*), and finally a form **mens-* with removal of the hiatus. But such a pre-form will account only for Ion. μείς, and not Att. μήν (gen. sg. μηνός) or forms from other dialects (cf. Lesb. gen. sg. μῆννος), which require a pre-Greek **mēns-* — thus without the syllabification and subsequent hiatus treatment Lindeman claims. (On the failure of Osthoff's Law to operate on **mēns-*, see the discussion by Peters, 1980:307ff.)

underlies ὄλλυμι 'destroy',[75] which Rix reconstructed as *$*h_3\underline{l}$-né-h_1-ti* (1970:94). The remodeling of this pre-form to the attested *νυ*-present is in fact somewhat unexpected, insofar as nasal presents built to roots ending in *$*$-h_1-* (a rare subtype) seem generally to be replaced with thematic formations rather than *νυ*-presents.[76] From this point of view, and ignoring, for the sake of argument, other evidence pointing to root-final *$*$-h_1-* (aor. ὀλέσαι etc.), as well as the fact that Harðarson (loc. cit.) provides a perfectly plausible analogical account for the *νυ*-present, a reconstruction *$*h_3\underline{l}$-né-h_3-ti* would account for ὄλλυμι in a more straightforward way — and this, in fact, is the source preferred by Lindeman (1997:54), who also notes that the same stem (but with the zero-grade form of the infix) is likely to be continued in the Hitt. *ḫalla-* of *ḫallaniya-* 'lay waste, ruin' (a form that in addition guarantees the initial laryngeal). So far as I am aware, however, Lindeman has not explained why such a form should not have turned up as Gk. †ἄλλυμι, under the assumption that initial *$*\underline{r}$-* in Greek should produce *a*-vocalism outside of Aeolic and Achaean. One presumes, based on the pattern of argumentation established by his approach to other cases (cf. on ἐρσήν [**3.5.**], ὄργυια [**3.6.**(ii)]), that recourse would be had to analogical influence from other parts of the paradigm showing *o*-vocalism (such as aorist, perfect, or future); but this possibility has already been countered by Rix (loc. cit.): "Eine analogische Übertragung der vollstufigen Wurzelform in den Präsensstamm ist wenig wahrscheinlich, da dabei ein totaler Allomorphersatz mit dem Ergebnis *ὀλέννυμι (cf. στορέννυμι, κορέννυμι) zu erwarten gewesen wäre".

4. According to the alternative view of the development of initial *$*\underline{r}$-* in Greek, the pandialectal result of absolute initial #RC- (following the loss of the initial laryngeal) is taken to be #ὰRC-.[77] This position would have the following consequences:

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[75]The only attested vocalism for this word — hence ὀλ- cannot be ascribed to an Aeolic treatment.

[76]See LIV 298, with reference to Harðarson (1993:223); cf. *$*g^{w}\underline{l}$-né-h_1-* → βάλλω 'hurl', *$*t\d{m}$-né-h_1-* ‡ τάμνω 'cut'.

[77]This widespread presumption (see e.g. Lejeune 1972:196) appears to correspond to Lindeman's actual view of the matter (cf. **2.4.**, ad init.; perhaps

(i) The interpretation of Hom. ὄρχαμος and Myc. *o-ka* presented above (**3.0.**) cannot be correct.

(ii) For the evaluation of Rix's Law, cases with initial *$*h_2\underset{\circ}{R}C$-* are not probative (as Lindeman and others have correctly noted), since these would coincide with the development of *$*\underset{\circ}{r}C$-* in any case. In contrast, however, initial *$*h_1\underset{\circ}{R}C$-* and *$*h_3\underset{\circ}{R}C$-* should both develop to #ἀRC- pandialectally (following the loss of the laryngeal), as opposed to Rix's claim that these sequences result in Gk. ἐRC- and ὀRC-, respectively.

Ad (i): this is unproblematic, as both of these difficult forms have viable alternative sources or interpretations that do not require initial *$*\underset{\circ}{r}$-* (e.g. *$*h_2org^h$-* or *$*sorg^h$-* for ὄρχαμος, /ōgā/ for *o-ka,* etc.).[78]

Ad (ii): the *$*h_1\underset{\circ}{R}C$-* and *$*h_3\underset{\circ}{R}C$-* forms already adduced (**3.5.**, **3.6.**) serve equally well here: again, these confirm the Rix's Law treatment, as against the implied claim that (in this case) *$*h_1\underset{\circ}{r}\underset{\smile}{u}$-é/ó-* would yield Att. †ἀρέσθαι/Ion. †ἄρέσθαι, *$*h_3\underset{\circ}{r}\hat{g}$-u-* would yield †ἀργυ-, *$*h_1\underset{\circ}{r}$s-ḗn* would yield †ἀρσήν/ἀρρήν (and with *a*-vocalism *exclusively*, vs. Lesb., Ion. [Hdt.] ἔρσην etc.), *$*h_3\underset{\circ}{l}$-n(u-)* would yield †ἄλλυμι, etc.

5. We may now summarize the results arrived at so far. Among the objections to Rix's Law that have been put forward, and leaving aside certain phonological misconceptions some of these objections incorporate, we identified as potentially most damaging the following pair of assertions about the development in Greek of Rix's Law input sequences *$*H\underset{\circ}{R}C$-*: first, since the laryngeal occurs in the position before a syllabic segment, it should regularly disappear, just as laryngeals disappear before vowels. Then, the resulting sequence *$*\underset{\circ}{R}C$-* should develop according to regular patterns of Greek sound change. Thus, to take the sole type of instance regularly cited in this connection by Lindeman and other critics of Rix's Law: if *$*\underset{\circ}{R}C$-* regularly develops to Gk. #ἀRC-, then it is impossible to

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also to that of Meid and Meißner, n. 26 above). Cf. recently Weiss (1998:39n14): "#*$\underset{\circ}{r}C$-* — assuming that PIE had any *r*-initial roots, which is uncertain — could only have given #*arC-*" [scil. in Greek]. n LIV s.vv. (note especially Kümmel 2000b), see also the following recent discussions: Hackstein 1998:227f., García Ramón 2000:65n5, Schmidt 2000, Jasanoff 2003:150 with n. 9, 166n39, 212 with n. 57.

[78] See Weiss loc. cit. (n. 77) and Vine 1998:698f. for discussion and references.

determine whether a surface #ə́RC- form derives from a Rix's Law "Laryngalwirkung" (e.g. **h₂R̥C-* > #ə́RC-), or from the ordinary treatment **R̥C-* > Gk. #ə́RC-.

This picture, however, has been shown to be seriously incomplete: it fails to consider, first, the two existing possibilities for the Greek treatment of pre-Greek sequences **R̥C-*; second, it ignores the distribution of these results by dialect; and third, it does not systematically address the behavior of sequences other than those involving **h₂-*. We have therefore investigated the following two approaches to the development of **R̥C-* in Greek, and the predictions that arise from them:

(i) **R̥C-* > #ə̀RC- in Aeolic and Achaean dialects and #ə́RC- elsewhere; if this is true, only the following are probative:

a. **h₁R̥C-* and **h₃R̥C-* in Attic-Ionic and West Greek: both should develop to #ə́RC- according to the above assumptions (vs. **h₁R̥C-* > #ɛ̀RC- and **h₃R̥C-* > #ə̀RC- by Rix's Law);

b. **h₁R̥C-* and **h₂R̥C-* in Aeolic and Achaean: both should develop to #ə̀RC- according to the above assumptions (vs. **h₁R̥C-* > #ɛ̀RC- and **h₂R̥C-* > #ə́RC- by Rix's Law).

(ii) **R̥C-* > #ə́RC- is pandialectal; if this is true, only **h₁R̥C-* and **h₃R̥C-* are probative, since these should both develop to #ə́RC- (vs. **h₁R̥C-* > #ɛ̀RC- and **h₃R̥C-* > #ə̀RC- by Rix's Law).

We have seen that *no matter which version of the development of *R̥C- in Greek is correct* (an issue on which it is unnecessary to take a position here), all of the predictions that would be made by critics of Rix's Law, according to the counterargument under consideration ("#*h₂*R̥C- [> #R̥C-] > Gk. #ə́RC- in any case", and its variants), turn out to be contradicted by the data. Instead, the crucial items — including at least some etymologically and morphologically unambiguous material, as well as forms reconstructed identically (or almost identically) by Rix and Lindeman — are attested in patterns that conform to the original statement of Rix's Law: thus Myc. and Arc. /Alki-/ (not †/Olki-/), pandialectal ἄρκτος (not Aeol. †ὄρκτος etc.), Att. ἐρέσθαι (not †ἀρέσθαι), ὄλλυμι (not †ἄλλυμι), etc. The overall result, therefore, is that once the full implications of the counterargument based on the development of **R̥C-* in Greek are weighed together with their logical consequences,

including consideration of non-Attic-Ionic dialect material, the relevant data force critics of Rix's Law into what may be characterized as a "no-win situation".

The remaining task is to consider an additional perspective injected by Meißner into the discussion, namely his critique of the existence of the two PIE roots **h₁er-* and **h₃er-*, as proposed by Rix and as used in several of his key examples (for the former, cf. ἔρχομαι [**2.1.**, **3.5.**]). It will be shown that, apart from difficulties with Meißner's argumentation, data from Mycenaean and Homer (hitherto not adduced in this connection) may provide additional evidence for the existence of two separate roots at a very early stage of Greek, and that even if only one of these roots actually existed at the level of PIE, at least one of the Mycenaean forms lends additional support to Rix's Law.

6.0. One of the most significant of Rix's proposals was the suggestion that the multitude of forms traditionally assigned to the "große Bewegungswurzel" PIE **er-* (cf. IEW 326ff.) should in fact be distilled into two originally distinct roots: those belonging to a root **h₁er-* 'zugehen auf', as in ἔρχομαι and Ved. *ṛcháti* (**2.1.**), and those descended from a root **h₃er-* 'aufspringen, losstürzen', as in ὄρνυμι 'raise'/ὄρνυμαι 'rise', Ved. *ṛṇóti* 'sets in motion', Lat. *orior* 'rise' and other forms (1970:92f., 100).[79] This suggestion has been widely accepted; perhaps not surprisingly, it appears in the influential compendium LIV (edited by Rix), which sports separate entries for **h₁er-* 'wohin gelangen, geraten' and **h₃er-* 'sich in (Fort-)Bewegung setzen'. Nevertheless, agreement is far from universal; indeed, a veritable cottage industry has arisen around the problem, such that a complete listing of discussions of it since 1970 is hardly possible.[80] Such a task would be rendered still more difficult by the fact that the problem intersects with another hotly-debated issue, namely the treatment of initial **h₃-* in Hittite: thus, Rix had originally

[79]As with ἔρχομαι/*ṛcháti,* the Rix's Law treatment of **h₃rC-* was justified by forms with morphologically secure zero grade: in addition to the *nu*-present ** h₃r-néu̯-* (ὄρνυμι), note also the *tó*-participle ** h₃r̥-tó-* (κονιορτός 'dust-cloud').

[80]In addition to the bibliographical references cited in LIV s.vv. (note especially Kümmel 2000b), see also the following recent discussions: Hackstein 1998:227f., García Ramón 2000:65n5, Schmidt 2000, Jasanoff 2003:150 with n. 9, 166n39, 212 with n. 57.

assigned Hitt. *arnuzi* 'move along (trans.), bring' to **h₃er-*; but other evidence suggests that initial **h₃-* is maintained in Hittite as *ḫ-*.[81]

6.1. The final section of Meißner's paper (1998a:40ff.) offers a detailed critique of Rix's proposal. His most cogent point begins with the observation that the alleged semantic differentiation between **h₁er-* 'move (forward)' (horizontally, as it were, cf. ἔρχομαι) and **h₃er-* 'rise, start up' (vertically, cf. ὄρνυμι, ὄρνυμαι) is undermined by the Greek data, where some forms with *e*-vocalism (e.g. ἔρετο· ὡρμήθη, ἔρσεο· διεγείρου Hsch.; ἔρνος 'shoot, sapling', with a meaning like ὄρμενος 'shoot, sprout', a noun that is formally identical to the aor. mid. ptcple. of ὄρνυμι) seem to belong semantically with ὄρνυμι. If, moreover, all of this material is to be accounted for by a single etymon (as would therefore seem attractive, on semantic grounds), that etymon, in view of the *e*-forms, cannot be "**h₃er-*," which could only produce forms with *o*-vocalism.

These points are well taken, and raise important considerations — as, indeed, had been seen before (see e.g. Peters 1993:398n105, ad fin., with references). But for followers of the Rixian dichotomy between **h₁er-* and **h₃er-*, such objections would be relatively unproblematic: these attestation patterns merely document the hypothesis that the two roots in question (and their reflexes) had influenced each other (partly as a result of inevitable phonological ambiguities) for some time, resulting in precisely such apparent overlaps — hence various situations of this sort described by LIV, e.g. Hitt. *arta* 'stood' assigned to **h₁er-* (238), despite its meaning; Ved. *ṛṇóti* 'sets in motion' with **h₃er-*, but "Teilweise wohl auch zu **h₁er-* und 1. **h₂er-* gehörig" (300n10); Ved. *ā́ra* 'ist wohin geraten' with **h₁er-*, but "[v]ielleicht auch Perfekt zu [1]*ar-* < **h₃er-*" (238n10); etc. (see again Kümmel 2000b, with discussion of the Indo-Iranian material in these terms).

6.2. Meißner's own solution, apart from an offhand suggestion that he does not pursue,[82] is far from compelling.

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[81] Among recent discussions, see e.g. Kimball 1987 and 1999:393f., Melchert 1987 and 1994:72, Rasmussen 1992, Ofitsch 1995, Lindeman 1997:46, 49f., Zeilfelder 1998:446 with n. 52, Tremblay 1999/2000:217, Oettinger 1995:216n16 (on Hitt. *arki-* 'testicle' and related forms, cf. Lindeman 1997:51n41, Katz 1998:207n65) and 2004.

Given, on the one hand, *e*-grade *nu*-presents like δείκνυμι, ζεύγνυμι, μείγνυμι etc., as opposed to the *o*-vocalism of forms with liquid stem of the type ὄρνυμι, στόρνυμι, θόρνυμαι etc., he proposes (updating a suggestion of Johannes Schmidt) that the latter forms owe their vocalism to a Greek *e*-umlaut rule, whereby (as a first approximation) *eR(R)u* > *oR(R)u*. While the rule seems at first glance to work rather well for *-νυμι* verbs — the prime *explanandum,* after all, being ὄρνυμι — it faces overwhelming obstacles in the form of numerous counterexamples outside this class, many of them (ἐρυθρός, πέρυσι etc.) noted and discussed by Meißner himself. Yet even the *-νυμι* verbs are not trouble-free: Hsch. θάρνυσθαι and θαρνεύεσθαι (vs. θόρνυμαι) compel Meißner to propose a complex reformulation of his original rule, leading to further problems that can safely be left aside here.[83]

Another approach via a single root **h₁er-*, and in this case prominently appealing to *o*-grade forms **h₁or-,* has recently been offered by Jasanoff (2003:212 with n. 57), whose analysis, it should be noted, is couched in very tentative terms. Jasanoff posits a complex development of competing archaic **h₁er-* and **h₁or-* forms, in which "the danger of homophony" would have favored "the maintenance and spread of *o*-grade", a process that eventually could have resulted in the creation, at some stage of the protolanguage, of a root (or "pseudo-root") "**or-*" — the entity that itself, in Jasanoff's view, would account for attested *o*-grade forms like Gk. ὦρτο, ὄρμενος, Lat. *orior,* and others. In a sense, however, the picture is a traditional Rixian

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[82] "I would not discount the possibility that *or- < *(h_x)o-h_1r̥- 'go upwards' or the like" (1998:40) — a possibility that one may safely discount. Among other problems, there is no evidence for a PIE particle **(H)o-* meaning 'upwards' or the like.

[83] We may note in passing that Meißner's presumed †ἔρνυμι (< **h₁r̥-néu̯-*) — the ultimate goal of his entire explanatory apparatus — is not actually of much help, as Meißner himself appears to recognize (1998a:42). As he notes, referring to forms with clearly secondary full grades (of the type δείκνυμι, ζεύγνυμι, etc., cf. above), "it is questionable whether it is legitimate to admit the large-scale replacement of zero grade by full grades". This is especially true here, as an inherited nasal present (with normal zero grade) is assured (unlike δείκνυμι etc.). Thus, *e*-vocalism in a form like †ἔρνυμι could only be explained via Rix's Law, since otherwise — depending on one's assumptions about the development of **r̥-* in Greek — such a zero-grade nasal present should result in Attic-Ionic †ἄρνυμι (**5.**(i)a.) or pandialectal †ἄρνυμι (**5.**(ii)).

one, with opposing *o*-grade and *e*-grade forms (e.g. *o*-grade root aorist middle in ὤρτο, ὄρμενος vs. *e*-grade middles in Ved. *ártа,* ptcple. *arāṇá-,* Gk. ἔρετο), and with the distinction between PIE **h₁er-* and **h₃er-* transposed to a late-IE contrast between **(h₁)er-* and a root-like entity "**or-*". Thus the PIE situation remains unclear, and the resolution of this problem, as Jasanoff writes, "must remain a task for the future".

6.3. Nevertheless, there is an additional Greek /er-/ ~ /or-/ alternation that may still usefully be considered in an evaluation of Rix's Law, and that seems never to have been addressed in that connection. I refer to a series of so-called "τερψίμβροτος compounds" (cf. Knecht 1946, more recently Meißner and Tribulato 2002:298f., 305f.), namely Myc. *e-ti-ra-wo* /Erti-lāwos/ (the τερψίμβροτος version of Gk. Λᾱ(ϝ)έρτης), *e-ti-me-de-i* /Erti-mēdehi/, beside *o-ti-na-wo* /Orti-nāwos/, cf. (with the same first member as *o-ti-na-wo*) Hom. Ὀρτίλοχος.[84]

The analysis of these forms naturally depends on the morphological (and, as we shall see, phonological) status of the first members of τερψίμβροτος compounds, which I have discussed in detail elsewhere (2004, with earlier references). The salient points for our purposes are these:

(i) As is generally agreed, the first members of archaic τερψίμβροτος compounds reflect *ti*-stem abstracts (still unassibilated in forms like Hom. Ὀρτίλοχος[85]); in Greek, the assibilated versions secondarily develop associations with sigmatic verb stems (such as *s*-future and *s*-aorist stems — cf.

[84]For full discussion of these forms, including justification of the interpretation of their first members as 'in Bewegung setzend, antreibend' et sim. (based on the root(s) of Gk. ὄρνυμι, ἔρετο etc.), see in detail García Ramón 2002. The alternative conception of *o-ti-na-wo* by Kazanskij (1998:32) in terms of /Ort^{h}i-nāu̯os/ 'he who guides the ship' interestingly arrives at a semantic interpretation close to that of García Ramón ('der die Schiffe antreibt [als Ruderer]'); but his etymological interpretation (cf. Gk. ὀρθός 'straight') is unappealing both for phonological reasons (Vine 1999:316n4) and in view of the phraseological evidence amassed by García Ramón. Even if it were shown to be correct, we would still be left with the unambiguous Ὀρτι- of Hom. Ὀρτίλοχος.

[85]It cannot be coincidental that the grandson of one of the Homeric figures so named has the later assibilated form Ὀρσίλοχος (textual and other details on both names in García Ramón 2002:190ff.). On unassibilated Hom. Ὀρτι- and Myc. *o-ti-* as archaisms, see Meißner and Tribulato 2002:306.

τερψίμβροτος 'delighting men' itself, beside fut. τέρψομαι 'will delight in' etc.).[86] In general, then, we will not be concerned here with the secondary layer of Greek sigmatic forms.

(ii) In keeping with the form in which plain *ti*-stems tend to be attested in IE languages, archaic τερψίμβροτος first members generally appear with zero grade (as opposed to secondary Greek sigmatic forms, with full grades related to other sigmatic stems; cf. again τερψίμβροτος beside fut. τέρψομαι etc.), as in RV *púṣṭi-gu-* [MN] (lit. 'cattle-rearing'), Myc. *qi-si-ja-ko* (n. 48 above). As original proterokinetic forms, *ti*-stems could in theory show full-grade/zero-grade root ablaut. In practice, however, both plain *ti*-stems and the first members of τερψίμβροτος compounds only rarely preserve full grade root forms (thus, like Ved. *púṣṭi°*, Myc. *qi-si°* just cited, also Ved. *gáti-* 'movement', like Gk. βάσις 'id.', etc.); it is often assumed, in fact, that zero grade was generalized for *ti*-stems at a late stage of the protolanguage (so e.g. Rix 1992:146).

(iii) The distribution of full grades is in fact severely limited in terms of root structure.[87] By far the best-attested full-grade τερψίμβροτος type is **CeH-ti°,* and **CeH-ti-* is likewise well-attested in plain *ti*-stems (including some of the same roots):

**CeH-ti° // *CeH-ti-*	
**deh₃-* 'give'	RV *dátivāra-* 'generous', Hom. *δωτ(ι)-[88] // Av. *-dāiti-* 'giving', OCS *blago-datĭ* 'grace'
**steh₂-* 'stand'	Myc. *ta-t(i)-* (/Stāt(i)-/, cf. Στησ(ι)-)[89] // Av. *stāiti-* 'resistance', OCS *po-statĭ* 'portion'
**(s)teh₂-* 'steal'	Myc. *ta-ti-qo-we-u* (/Tāti-g^{w}oṷeus/) // OIr. *táid,* OCS *tatĭ* 'thief'

[86]A somewhat more complex "polygenetic" approach to τερψίμβροτος compounds has been developed by Dunkel (1992); his system likewise acknowledges, in any case, the generally secondary character of the Greek sigmatic forms.

[87]Further details on a number of the forms listed here and below in Vine 2004.

[88]E.g. Δωτικλέης vel sim., implied by the Kurzname Δωτώ (Σ 43), cf. Knecht 1946:11.

[89]Myc. *ta-ta-ke-u* (n. 48 above).

**CeH-ti̯o*	
**gʷeh₃-* 'nourish'	Hom. *βωτιάνειρα* 'nourishing men'
**gʷeh₂-* 'step'	Hom. *βητ(ι)-* (*βητάρμων* 'dancer', cf. MN *Ἀναβησίνεως*)

**CeH-ti-*	
**meh₁-* 'measure'	Skt. *māti-* 'measure', Hom. *μῆτις* 'counsel', Lat. *mēti(or)* 'measure', OE *mǣþ* 'measure'
**dʰeh₁-* 'put'	Av. *δāiti-* 'stopping', Lat. *fēti(ālis)* 'referring to the fetial (priestly term)', Go. *ga-deþs* 'deed',[90]
**speh₂-* 'thrive'[91]	RV *sphātí-* 'fattening', Gmc. **spōði-* (OHG *spuot,* OE *spēd* 'success')
**peh₃-* 'drink'	Lat. *pōtiō* 'drink', [Pi.+] *ἀνάπωτις* ~ [Hdt.+] *ἄμπωτις* 'ebb'

and a number of isolated forms, e.g. to **reh₁-* 'give' (Ved. *rātí-* 'gift', Av. *rāiti-* 'generosity'), **bʰeh₂-* 'shine' (Skt. [BhP] *bhāti-* 'light'), **seh₂-* 'bind' (Av. *hāiti-* 'section'), **seh₁-* 'sow' (OE *sǣd* 'seed', Go. *mana-seþs* 'mankind').

(iv) No other full-grade types are found with any frequency among *τερψίμβροτος* compounds; and in general, much as in (iii), the root structure of isolated full-grade forms is paralleled in plain *ti*-stems. Thus are found, e.g.:

a. **Ces-ti̯o*: **nes-ti̯o* (Myc. *Ne-ti-ja-no* /Nesti(i̯)-ānōr/, cf. Thessal. Νέσσανδρος)[92]

a'. **Ces-ti-*: **u̯es-ti-* (Lat. *uestis,* Go. *wasti* 'clothing', Skt. [gramm.] *vasti-* 'fringe')[93]

b. **CeN(H)-ti̯o*: perhaps **h₁enh₃-ti̯o* in *ἐνοσι-* (*ἐνοσίχθων* etc.; Janda 1999), although zero grade is also possible (see Janda 1999:187)

b'. **CeN(H)-ti-*: **ĝenh₁-ti-* (Av. *frazaiṇti-* 'posterity, offspring', Gmc. **kindi-* [Go. *kindins* 'governor' etc.],

[90]Here perhaps also Lyd. *ta(a)c-* 'votive tablet/offering' (Melchert 1997:34ff.).

[91]For the reconstruction of this root with **-h₂-*, see Jasanoff 2003:108f.

[92]Cf. also García Ramón 2002:186 on Myc. *ka-sa-no* /Kassānōr/ < **kas-ti-ānōr*.

[93]Cf. the related amphikinetic formation in Toch. B *yesti* < **u̯és-toi̯* (Malzahn 2002/03).

OCS *zętiˇ* 'son-in-law'); **men-ti-* (Skt. [gramm.] *manti-* 'thinking', Go. *ana-minds* 'supposition')[94]

(v) Reminiscent of **u̯es-ti-* ((iv)a'. above) is a series of *ti*-stems of the shape **u̯eK-ti-* (K = velar or palatal), see Vine 2004:373n46 on **u̯eĝʰ-ti-*, **u̯ek̂-ti-*, and **u̯ek-ti-*. Together with **nes-ti*° ((iv)a. above), this material suggests that "samprasāraṇa" roots favored full grade **ReC-ti-* over zero grade **R̥C-ti-*; cf. Lat. *messis* 'harvest' < **met-ti-*, among others of this type.[95]

(vi) It hardly needs to be added, finally, that a number of isolated *ti*-stems built to *CeC-* roots show full grade, as is entirely expected (cf. verbal adjectives in **CeC-tó-* for ideal ***CC-tó-*): **ked-ti-* (Lat. *cessim* 'gradually', *necesse* 'unavoidable'), **gʰebʰ-ti-* (ON *gipt* 'gift'), etc.

6.4. The above survey shows that for archaic Greek τερψίμβροτος first members of the shape /Erti°/ (Myc. *e-ti-*) and /Orti°/ (Myc. *o-ti-*, Hom. Ὀρτι-), there is no justification for morphological interpretations as full grades: roots of the structure **HeR-* fall into none of the categories that typically attest full-grade *ti*-formations (whether in simplex or compound types). Since the first members of archaic τερψίμβροτος compounds regularly show zero grade (**6.3.**(ii)), the most straightforward analysis of the perfectly parallel and contemporaneous formations /Erti°/ and /Orti°/ in early Greek[96] would thus operate with zero-grade sources. This, in turn, suggests that both **h₁er-* and **h₃er-* may indeed need to be reconstructed, and further that /Erti°/ continues **h₁r̥̄-ti-* (by Rix's Law) and that /Orti°/ continues **h₃r̥̄-ti-* (by Rix's

[94]A pattern with a certain fortune in Indo-Iranian: see Liebert (1949:72ff., 108f.) on RV *tantí-*, *ránti-*, VS *á-hanti-*, etc.

[95]According to a theory developed by J. Schindler (cf. 1994:398f.; further Jasanoff 2003:109n40, Vine 2004:367, 373n46, and the critical treatment by de Vaan 2004), "unexpected" full grades (and lengthened grades) in primary nominal formations may in some cases be related to corresponding acrostatic formations for the verbal roots in question — as possibly here, for example, in the case of **u̯es-ti-* (cf. again Malzahn 2002/03); note also Myc. *me-ti-ja-no* /Mēs-ti(i̯)-ānōr/ < **mēd-ti*°, cf. μήδομαι 'consider' beside μέδομαι 'be mindful of, devise', μῆδεα (neut.) 'advice'.

[96]On the morphological parallelism: cf. ὦρτο, ὄρμενος beside ἔρετο, ἔρσεο (**6.2.**).

Law).[97] As often, it would be possible to claim that one or both of these forms has been influenced by a related full grade — e.g. /Erti°/ (for the "anti-Rix" phonologically regular †/Orti°/ or †/Arti°/ < *$h_1\underset{\circ}{r}$-ti-*) under influence from the (poorly-attested) full-grade middle *$h_1$ér-to* (cf. ἔρετο). But this would be a highly unattractive assumption for the first member of isolated (NB: unassibilated) τερψίμβροτος compounds, a recessive morphological type that is everywhere attested in the form of relics, and in this case found (not surprisingly) in the earliest Greek available to us.

But let us suppose that Jasanoff, Meißner and others are correct in claiming that only *$h_1$er-* is to be reconstructed for PIE (e.g. with the forms showing *o*-vocalism somehow derived, following Jasanoff, from an invariant late IE pseudo-root or neo-root "*or-", ultimately traceable to *o*-grade *$h_1$or-* forms). Even in this case, however, Rix's Law is inescapable, no matter which version of the development of initial *$\underset{\circ}{r}$- in Greek one prefers. In terms of the "anti-Rix" predictions explored above: if preconsonantal *$h_1\underset{\circ}{r}$- > *$\underset{\circ}{r}$-, and if this in turn develops to /or-/ in the dialect continuum underlying Mycenaean Greek, then Myc. /Erti°/ < *$(h_1)\underset{\circ}{r}$-ti-* has no explanation except by Rix's Law; and if, instead, *$\underset{\circ}{r}$- > /ar-/ (in Mycenaean, as everywhere else), then *neither* /Erti°/ *nor* /Orti°/ (the latter including Hom. Ὀρτι-) can be explained in the way that is claimed by Lindeman and other critics of Rix's Law.

Thus the early Greek data from τερψίμβροτος compounds eliminates the "*$h_1$er-/*$h_3$er- problem" as a counterargument to the liquid version of Rix's Law, since Myc. /Erti°/ and Myc.-Hom. /Orti°/ force critics of Rix's Law into the same impasse demonstrated above (**5.**). Yet more is at stake than a sub-part of a single phonological process; for the validity of Rix's Law can only be interpreted as confirmation for the "triple representation" model of laryngeal vocalization in Greek (cf. n. 8 above), an issue of vastly greater scope and significance for Greek and Indo-European phonological history.[98]

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[97] García Ramón is non-committal as to whether these forms derive from *$h_1$er-*, *$h_2$er-*, or both (2002:185 with n. 11), and does not comment on their original vocalism; Dunkel explicitly ascribes a zero-grade source to Myc. *o-ti-* /Orti°/ (1992:222).

[98] Earlier versions of this paper were presented at the 19th East Coast Indo-European Conference (Univ. of Georgia, June 2000), at the University of

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